
Refinement Buys Intelligibility, Search Buys Identity: What Test-Time Compute Buys in Masked-Diffusion TTS

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Abstract

A diffusion language model buys sequential computation twice: depth, paid once in parameters, and denoising steps, rented per utterance at serve time. We ask whether the rented kind helps every capability equally. We train an iso-parameter width–depth grid of 15 masked-diffusion codec TTS models (3 seeds, 19–133 M non-embedding parameters) on 2,000 h of Emilia-EN and sweep the refinement budget T at inference only, evaluating cross-sentence zero-shot synthesis on 400 items from 174 held-out speakers. Our central result is model-free and referenced to floors we *measure* rather than to zero: the word error of an ASR system on the real recordings, and the speaker similarity of a codec round trip. Against those floors, refinement from $T=1$ to $T=16$ closes 86.2% of the reachable intelligibility range but only 46.4% of the reachable identity range, a $1.86\times$ asymmetry that holds between $1.68\times$ and $1.86\times$ under every monotone reparameterisation of the error we tried. We also report the pre-registered power-law fit, and we report that the exponent contrast it yields ($\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}} = 0.110$) changes sign under those same reparameterisations: it is a property of the error coordinate, and we therefore decline to build the paper on it. The limit is on what *refinement* buys, not on what inference buys. At matched NFE, best-of- K search moves speaker similarity where refinement cannot, and 4 speaker-encoder families outside the selector’s lineage agree, with per-item win rates of 64.6–79.0% and every interval above one half. Depth and steps are *not* interchangeable: the substitution form $B(dT^\kappa)^{-\beta}$ fits worse than the separable form by $\Delta\text{AICc} = +69.3$. Finally, 62% of the remaining identity gap belongs to the codec, not the model — so most of what identity has left to gain is not the generator’s to win.

1 Introduction

A diffusion language model can buy sequential computation twice. It *owns* depth: d transformer blocks, paid once in parameters and then on every forward pass forever. It *rents* refinement steps: T denoising passes over the same sequence, paid per utterance and adjustable at serve time. An autoregressive model has only the first lever; its serial cost per token is fixed by the architecture the moment training ends. If rented serial compute substitutes for owned serial compute, a deployment can trade parameters for steps after the fact. That trade is available only for the capabilities that steps improve.

Zero-shot text-to-speech is a clean probe: intelligibility is compositional (the right phonetic content in the right order, read out by word error rate) while speaker identity is representational (timbre carried from a short prompt, read out by SIM-o). The two need not respond to the same knob.

We train one iso-parameter width–depth grid of masked-diffusion codec language models and then sweep T at inference only, which yields a full (w, d, T) response surface per metric at zero additional training cost. Four hypotheses were pre-registered before the first run: that refinement is metric-selective (H-D2, primary), that steps and depth exchange at a rate κ (H-D3), that shape matters at fixed parameter count (H-D1), and that small budgets predict large ones (H-D4). Table 8 gives every verdict, including the failures. Eq. (1) states the two fitted forms: one separable in width, depth and steps, one in which steps substitute for depth through the product dT^κ .

$$\begin{aligned} \text{separable: } \text{err}_m(w, d, T) &= E_m + A_m w^{-\alpha_m} + B_m d^{-\beta_m} + C_m T^{-\tau_m} \\ \text{substitution: } \text{err}_m(w, d, T) &= E_m + A_m w^{-\alpha_m} + B_m (dT^{\kappa_m})^{-\beta_m} \end{aligned} \quad (1)$$

The pre-registered test is supported in the coordinate it declared: $\Delta\tau = 0.110$, run-level bootstrap CI $[0.092, 0.131]$, excluding zero. But τ is only guaranteed invariant to *affine* rescalings of the error, which is what our pre-registration checked, and $\Delta\tau$ changes sign under monotone non-affine ones — $\log(1 + \text{err})$ and $\sqrt{\text{err}}$ among them (§4.2). A quantity whose sign depends on whether one writes WER or $\sqrt{\text{WER}}$ describes a coordinate, not a system. We therefore restate the finding in a form that does not move: against floors we measure, refinement closes 86.2% of the reachable intelligibility range and 46.4% of the reachable identity range. Depth and steps, meanwhile, do not exchange at a rate one could design around: the substitution form fits *worse* than the separable one ($\Delta\text{AICc} = +69.3$ for WER, $+71.3$ for SIM-o) and κ runs to its bound, so no single ratio holds across the grid. Extra inference compute is a targeted instrument, not a substitute for architecture.

Contributions. ❶ The first per-metric (w, d, T) scaling surface for speech generation: 75 points per metric from 45 trained models, released with the training and analysis code. ❷ A floor-referenced, model-free measurement of what refinement buys per capability (86.2% versus 46.4% of the reachable range), together with the scope conditions under which the usual exponent-based statement of the same effect fails — a reparameterisation, a fitted range, and a goodness-of-fit test its own criterion rejects. ❸ A per-axis account of what moves speaker identity, measured against a codec ceiling: training compute and test-time selection move it where refinement does not, while longer prompt context and rate matching buy nothing, with the selection result confirmed by 4 encoder families outside the selector’s lineage.

2 Related work

Masked diffusion language models are trained at scale [Nie et al., 2025, Sahoo et al., 2024, Austin et al., 2021], but their scaling is reported against a single likelihood-shaped objective, which cannot expose a per-capability difference. The masked-generative recipes that made codec-token TTS practical [Borsos et al., 2023, Chang et al., 2022, Wang et al., 2024] fix a step budget by engineering judgement and report one operating point; none maps error against steps as a law. Autoregressive codec TTS [Wang et al., 2023] has been scaled, but in total parameters only [Ye et al., 2025], which separates neither width from depth nor either from steps. Classical scaling work aggregates the same way [Kaplan et al., 2020, Hoffmann et al., 2022]. Test-time scaling is well studied in reasoning [Snell et al., 2024], where the payoff is a single correctness metric. Whether extra inference compute helps different capabilities *differently* is, to our knowledge, untested in generative speech.

3 Method

Backbone and objective. The backbone is a bidirectional (non-causal) pre-LN transformer over Mimi [Défossez et al., 2024] codec tokens at 12.5 Hz, using the first 8 residual codebooks (2048 entries each, plus MASK and PAD). It is parameterised by **width** w and **depth** d , with head dimension 64, GELU MLP ratio 4, RMSNorm, RoPE and no weight tying, so the non-embedding parameter count is $N = 12dw^2$. One embedding table per codebook is summed at the input and one linear head per codebook reads out. Training follows SoundStorm-style [Borsos et al., 2023] coarse-to-fine masking: per example we draw a level $l \sim \mathcal{U}\{1..8\}$ and a ratio $t \sim \mathcal{U}(0, 1]$; levels below l stay visible, level l is masked i.i.d. with probability t , and levels above l are entirely MASK. The loss is cross-entropy on the masked cells of level l only. A phoneme prefix and a 3-second prompt of codec tokens are always visible. The model has no speaker encoder, so the backbone is the only identity

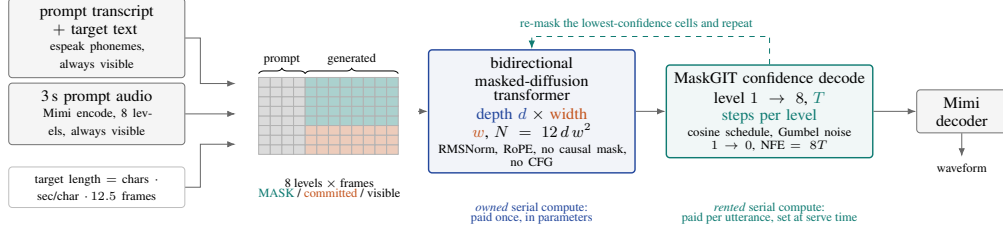


Figure 1: *The two ways a masked-diffusion TTS model buys sequential computation.* Phonemes and a 3-second prompt are always visible; the generated region of the $8 \times \text{frames}$ token grid starts fully **masked**. The backbone spends **depth** — *owned* serial compute, fixed at training time — and the sampler spends **T refinement steps per codebook level** — *rented* serial compute, chosen at serve time. One training grid over (w, d) therefore yields a whole (w, d, T) surface at inference. This paper asks whether the rented kind helps every capability equally.

83 pathway, and it is trained without classifier-free guidance, which would otherwise confound the step
84 axis. The target length is fixed before generation from the text length and a corpus character rate
85 (0.0602 s/char).

86 **Data and task.** All models are trained on a 2,000 h speaker-stratified subset of Emilia-EN [He et al.,
87 2024] covering more than 4,000 speakers, with 200 held out entirely. Evaluation is cross-sentence
88 zero-shot synthesis — prompt and target are *different* utterances from the same held-out speaker,
89 so copying the prompt’s content does not score — on 400 items from 174 unseen speakers, targets
90 4–15 s. Two measured floors anchor every number we report: the word error Whisper-large-v3 makes
91 on the real recordings of those items (0.0345), and the speaker similarity of a Mimi round trip of
92 the real target with no model in the loop (0.5554). Neither is reachable by a better generator, and
93 referencing gains to zero instead of to these floors is what makes the two metrics look incomparable.

94 **Frozen sampler.** Inference is MaskGIT-style confidence decoding [Chang et al., 2022], run level
95 by level with exactly **T refinement steps per level**, so $\text{NFE} = 8T$. A cosine schedule sets how
96 many cells remain masked after each step, tokens are drawn at temperature 1.0, and confidences are
97 log-probabilities plus Gumbel noise annealed $1 \rightarrow 0$; committed cells stay committed. The sampler
98 is identical for every config, seed and T , and generation noise is keyed so that T is the only difference
99 between surface points of one run.

100 **Design and statistics.** The grid holds 15 configs: three iso- N budgets $\times$ five aspect ratios, each
101 trained at 3 seeds for 30,000 steps at effective batch 256 with μP [Yang et al., 2021] (base width
102 256, base LR 0.004, transfer checked by five-point LR sweeps). We sweep $T \in \{1, 2, 4, 8, 16\}$ at
103 evaluation only, which gives 75 surface points per metric from 45 trained models. The metric stack is
104 WER (Whisper-large-v3 [Radford et al., 2023], greedy), SIM-o (WavLM-large [Chen et al., 2022])
105 against the *original* prompt waveform, following Anastassiou et al. [2024]), a frozen degenerate-
106 output rule, and UTMOS [Saeki et al., 2022]. The error transforms are $\text{err}_{\text{WER}} = \text{WER}$ and
107 $\text{err}_{\text{SIM}} = 1 - \text{SIM-o}$, computed over all items including degenerates. We fit both forms of Eq. (1) by
108 weighted non-linear least squares with 32 multi-starts. The intervals on $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}}$, κ and
109 $\Delta\rho = \alpha_{\text{SIM}}/\beta_{\text{SIM}} - \alpha_{\text{WER}}/\beta_{\text{WER}}$ are percentile intervals from a bootstrap of 2000 replicates that
110 resamples at the *run* level and recomputes every T point. The run level is the right unit because the
111 five T -points of one run share a trained model, so resampling surface points would understate their
112 correlation.

113 4 Results

114 4.1 What refinement buys, measured against floors

115 The comparison that matters is not how much error each metric sheds — WER and $1 - \text{SIM-o}$ have
116 no common unit — but how much of the error that is *reachable at all* each one sheds. Table 1 reports
117 that fraction directly from the measurements, with no fitted model. Going from one-step decoding to
118 the deployable $T=16$ default, refinement closes 86.2% (CI [82.8, 89.1]) of the distance from $T=1$ to
119 the ASR floor, and 46.4% (CI [45.0, 47.8]) of the distance to the codec ceiling: a ratio of $1.86 \times$ (CI

[1.82, 1.90]). Refinement very nearly exhausts what is available to it on intelligibility while leaving most of the identity range untouched.

Two properties make this the statement we build on. First, it is stable in T : we extended every run to $T=64$, and over that $16\times$ range of refinement budget the ratio stays between $1.85\times$ and $1.96\times$ while $\Delta\tau$ halves (§4.2). Quadrupling the budget past the deployable default takes intelligibility from 86.2% to 88.4% but identity only from 46.4% to 47.9% (CI [46.6, 49.2]): refinement plateaus against the codec ceiling at roughly half of what identity has left to gain, and buying more of it does not help. Second, it is stable in the coordinate: being a ratio of differences along one axis, a monotone reparameterisation moves numerator and denominator together, so across $\log(1 + \text{err})$, $\sqrt{\text{err}}$ and err^2 the ratio stays between $1.68\times$ and $1.86\times$ where $\Delta\tau$ changes sign.

The grid resolves shape differences well above seed noise: within each budget the spread of config means at $T=16$ is at least twice the pooled seed standard deviation, so the pre-registered power criterion passed in all three budgets.

4.2 The same effect as a fitted exponent, and why we do not lead with it

Our pre-registration declared a power-law surface (Eq. 1) and a primary test on its step exponents. That test passes: $\tau_{\text{WER}} = 0.836$ (CI [0.825, 0.851]) against $\tau_{\text{SIM}} = 0.726$ (CI [0.709, 0.744]), giving $\Delta\tau = 0.110$, CI [0.092, 0.131], excluding zero (H-D2). Three scope conditions bound what it means, and we state them because each one weakens a claim we would otherwise have made.

❶ *Coordinate*. τ is invariant to affine rescalings of the error, which is what we verified before running; it is not invariant to monotone non-affine ones. Refitting the identical measurements under $\log(1 + \text{err})$, $\sqrt{\text{err}}$ and a cap at $\text{err} = 1$ sends $\Delta\tau$ *negative* in 3 of the reparameterisations we tried (Table 2). Its sign is a fact about the coordinate. ❷ *Range*. τ summarises curvature over the window it is fitted on, and is not an identified asymptotic exponent. Extending every run to $T=64$ and refitting on nested windows, $\Delta\tau$ falls from $+0.2191$ over $T \leq 8$ to $+0.1102$ over the declared $T \leq 16$ and $+0.0660$ over $T \leq 64$: it roughly halves each time the window doubles. The declared value is a local slope at the edge of the declared grid. ❸ *Fit*. Judged by the weighted criterion the fit itself minimises, the declared form is rejected as a generative model: $\chi^2/\text{dof} = 14.3$ for WER and 8.8 for SIM-o. We use it as a compact description of curvature and read no mechanism from its parameters. The declared interval also holds the $1/\text{SE}^2$ weights fixed at their observed-data values; resampling the weights as well widens it to [0.0349, 0.1640].

None of this touches §4.1, which is model-free. It is why §4.1 is the result we lead with.

Table 1: *What refinement buys, as a fraction of what is reachable*. Each cell is the share of the distance from one-step decoding to a *measured* floor — the ASR word error on the real recordings (0.0345) and the speaker similarity of a codec round trip (0.5554) — that refinement closes by step budget T . Intervals are percentile CIs from a 2000-replicate bootstrap over the 45 runs. No model is fitted. The final column is the ratio of the two shares; it is a ratio of differences on a common axis and so, unlike $\Delta\tau$, keeps its size under monotone reparameterisation of the error ($1.68\times$ – $1.86\times$ across the maps in Table 2).

T	fraction of reachable range closed		
	intelligibility (WER)	identity (SIM-o)	ratio
2	32.2% [29.7, 34.4]	20.0% [19.1, 20.9]	$1.61\times$
4	69.8% [66.0, 73.1]	35.7% [34.3, 37.0]	$1.96\times$
8	81.8% [78.3, 84.8]	43.5% [42.1, 44.9]	$1.88\times$
16	86.2% [82.8, 89.1]	46.4% [45.0, 47.8]	$1.86\times$
32	87.9% [84.9, 90.5]	47.5% [46.2, 48.8]	$1.85\times$
64	88.4% [85.3, 91.0]	47.9% [46.6, 49.2]	$1.85\times$

4.3 Robustness of the asymmetry

The selectivity is not an artefact of scale, training compute or the metric stack (Table 3). A fourth parameter budget (276 M, ten further runs) gives $\Delta\tau = +0.1095$, CI [0.0951, 0.1241]; retraining three configurations $3\times$ longer gives $+0.1767$, CI [0.0789, 0.2854]; a different ASR gives $+0.1283$

Table 2: *The exponent contrast is a property of the error coordinate.* The same 75 measurements, refitted after a monotone reparameterisation of both errors. τ is invariant to the affine map, as our pre-registration assumed; under the non-affine maps $\Delta\tau$ changes sign, while the model-free ratio of Table 1 does not. This is why the paper’s claim is stated as a share of a reachable range rather than as an exponent difference.

error coordinate	τ_{WER}	τ_{SIM}	$\Delta\tau$	gap-closed ratio
identity (<i>pre-registered</i>)	0.8362	0.7260	+0.1102	$1.86\times$
$\log(1+\text{err})$	0.6665	0.6913	-0.0248	$1.85\times$
$\sqrt{\text{err}}$	0.4325	0.6842	-0.2517	$1.68\times$
err capped at 1	0.7093	0.7260	-0.0167	—
err squared	1.5613	0.8122	+0.7491	$1.82\times$
affine $3+2*\text{err}$ (v1.0 check)	0.7600	0.7249	+0.0351	—

155 and unbounded log-amplitudes +0.1260. Extending T to 64 leaves WER still falling past the grid
 156 edge ($0.1516 \rightarrow 0.1294$), so the gain measured at $T=16$ is a lower bound.

Table 3: $\Delta\tau$ survives variation in scale, training compute and the measurement stack. Each row refits the surface with one factor changed: a fourth parameter budget, $3\times$ training compute, a different ASR, and an unbounded reparameterisation of the fitted amplitudes. Every interval excludes zero.

variation	what changes	$\Delta\tau$	95% CI
none (75 points, 45 runs)	—	0.110	[0.092, 0.131]
fourth budget, 276 M	10 further runs	+0.1095	[0.0951, 0.1241]
$3\times$ training compute	3 configs at 90k steps	+0.1767	[0.0789, 0.2854]
ASR swapped	Whisper-medium.en	+0.1283	[0.1064, 0.1492]
unbounded amplitudes	$\log A, \log B, \log C$ free	+0.1260	[0.1043, 0.1474]

157 **Depth and steps do not substitute.** The substitution form of Eq. (1) fits the surface worse than
 158 the separable form: $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$ for WER and $+71.3$ for SIM-o, where the
 159 pre-registered claim (H-D3) required $\leq +4$, and κ sits at its bound in every fit. The honest reading is
 160 narrower than "no exchange rate exists". A global factor dT^κ is a strong constraint — it demands
 161 that one ratio hold at every depth and every step count — and it is that constraint the data reject. The
 162 separable fit still implies a *local* rate, $\kappa \approx \tau/\beta = 0.41$ for WER, valid near the operating point and
 163 nowhere claimed to extend beyond it. What does not survive is the design rule: one cannot trade
 164 depth for steps at a fixed rate across the grid.

165 **Shape matters, but the anisotropy ratio is not identified.** At $T=16$ the two-axis model beats the
 166 N -only model for both metrics ($\Delta\text{AICc} = -15.5 / -5.5$ against a -4 threshold), so error at fixed
 167 parameter count depends on how those parameters are shaped. The direction of that dependence does
 168 not resolve. The width amplitude saturates its pre-registered upper bound in every fit, so only the
 169 product $A w^{-\alpha}$ is identified and α is not, which leaves the anisotropy ratio $\rho = \alpha/\beta$ undetermined
 170 and the secondary pre-registered claim (H-D1) not supported.

171 **Depth** is identified, and it separates the capabilities sharply ($\beta_{\text{WER}} = 2.040$ versus $\beta_{\text{SIM}} = 0.553$):
 172 moving from the shallowest shape to the best depth is worth 40.2, 16.3 and 7.3 WER points across
 173 the three budgets, while SIM-o varies by under 0.065. The depth optimum is interior only in the
 174 smallest budget; in the two larger ones the argmin sits at the deepest shape tested, so those are lower
 175 bounds, not interior optima.

176 **Extrapolation and degenerate outputs.** Fitting the two smaller budgets and predicting the largest
 177 gives MAPE 2.4% for SIM-o against 2.2% for the N -only model: identity extrapolates across scale,
 178 within the 15% we pre-registered, but no more accurately than parameter count alone. WER does not
 179 extrapolate (20.2% versus 20.4%). Degeneracy falls monotonically with T , from 40.4% at $T=1$ to
 180 0.1% at $T=16$: most of what a one-step budget costs is outright failure, not degraded speech.

181 **Serving corollary.** Per-layer latency is flat in width: measured at batch 1, c_{layer} is 0.493–0.514
 182 ms for $w \in [256, 1152]$, because the device is launch-bound at this scale. Serial latency is therefore
 183 $8Td_{c_{\text{layer}}}$, so at fixed parameter count width is free and a latency budget constrains only the product

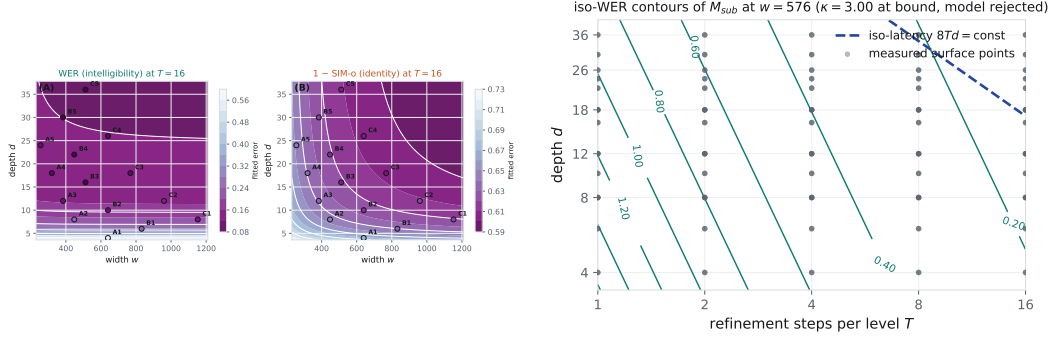


Figure 2: *Left*: iso-error contours in (w, d) at $T=16$. The two capabilities do not share a preferred shape, but the width amplitude saturates its bound, so the anisotropy ratio ρ is not identified. *Right*: iso-WER contours in $(\log T, \log d)$: a single substitution factor $d T^\kappa$ cannot reproduce both the depth ordering at fixed T and the T ordering at fixed depth.

184 $T d$. How to divide that product between depth and steps is not determined here. The pre-registered
 185 rule, buy depth before steps, is refuted: it holds at 5.9% of the 40 latency budgets we test, against a
 186 required 80%. The reverse ordering is a boundary effect of a grid censored at $T=16$, not a finding.

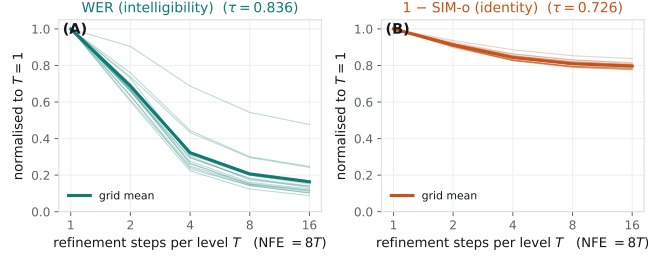


Figure 3: *Test-time scaling is metric-selective*. Error versus refinement steps for all 15 configs, each normalised to its own $T=1$ value; thick lines are grid means. (A) WER error falls 0.952 in absolute terms over $T=1 \rightarrow 16$ and is still falling at $T=16$. (B) SIM-o error falls 0.161 over the same range, $5.9\times$ less. Same models, same frozen sampler, same items — only T differs.

Table 4: *Fitted exponents* on the 75-point surface (M_{sep} for α, β, τ ; M_{sub} for κ), with 95% run-level bootstrap intervals on the step exponents. The primary result is $\Delta\tau = 0.110$, CI [0.092, 0.131], which excludes zero. κ sits on its bound in both rows and the substitution form is worse than the separable one by $\Delta\text{AIC}_{\text{sub-sep}} \gg +4$, so the depth-step exchange rate has no finite estimate. The negative $\Delta\text{AIC}_{\text{full-N}}$ says that shape matters at fixed N .

Metric	α (width)	β (depth)	τ (steps)	κ	$\Delta\text{AIC}_{\text{sub-sep}}$	$\Delta\text{AIC}_{\text{full-N}}$
WER	1.574	2.040	0.836 [0.825, 0.851]	3.000 [†]	+69.3	−15.5
SIM-o	0.919	0.553	0.726 [0.709, 0.744]	3.000 [†]	+71.3	−5.5

[†] at the fitted bound, i.e. not identified; $\Delta\tau = 0.110$, CI [0.092, 0.131].

187 **Training compute is the largest identity lever, and has not saturated.** Retraining three con-
 188 figurations to 90k steps raises SIM-o by +0.0827 (CI [0.0733, 0.0908], 9 runs, all positive), and a
 189 further doubling to 180k adds +0.0155 (CI [0.0064, 0.0243]) at a WER cost inside the ± 2.0 -point
 190 guardrail. Across {30k, 90k, 180k} steps on one configuration SIM-o runs $0.4081 \rightarrow 0.4807 \rightarrow$
 191 0.4962 . Extrapolating the first interval log-linearly predicts 0.5264 at 180k against 0.4962 observed,
 192 a residual of -0.0302: the axis is decelerating but still paying, and per unit it buys more identity than
 193 any inference-time intervention we tested.

194 **What buys identity, measured against a ceiling.** If refinement is an intelligibility knob, what
 195 moves identity? We put every candidate lever on one scale and read it against a fixed bound. A

Table 5: *What buys speaker identity.* Absolute SIM-o gain per axis, set against the 0.0747 of headroom between our best system and a codec round-trip of real audio. The headroom is a WavLM-large quantity, so every row is reported under that same encoder (final column); the cross-encoder agreement evidence for the search row is separated out into Table 7 rather than mixed into this column. Training compute and test-time selection buy the most. Guidance buys a little and charges for it in WER; rate matching and longer prompts buy nothing.

axis	change	Δ SIM-o	cost	scorer
training compute	30k $\rightarrow$ 90k steps	+0.0827	3 $\times$ training	WavLM-L
(further)	90k $\rightarrow$ 180k	+0.0155	2 $\times$ again	WavLM-L
test-time search	best-of-8 vs $T=16$	+0.0365	+0.0425 WER, 8 $\times$ NFE	WavLM-L
parameters	20 M $\rightarrow$ 276 M	+0.0531	14 $\times$ parameters	WavLM-L
guidance	$\gamma = 0.5$	+0.0121	+2.73 WER points	WavLM-L
rate matching	per-item duration	+0.0044	free	WavLM-L
prompt context	3 s $\rightarrow$ 9 s	none	+71.7 WER points	WavLM-L

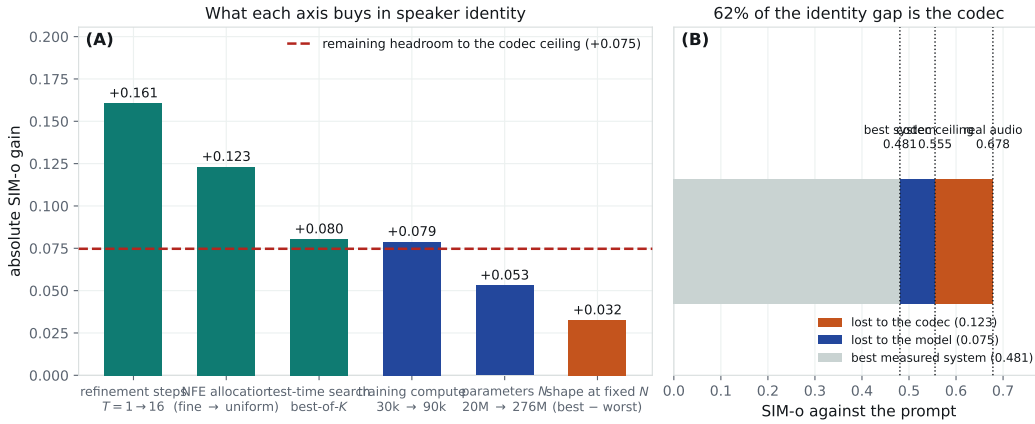


Figure 4: *Identity is mostly not the model's to win.* (A) what each axis buys in speaker similarity, against the headroom that remains to the codec ceiling; bars are coloured by which lever they are (rented at serve time, owned in training, shape). Refinement's bar is the largest only because it is measured from one-step decoding, which is not an operating point anyone ships; from the deployable $T=16$ default it buys nothing further, and test-time search is what moves identity from there. (B) of the distance between real audio and our best system, 62% is lost in the codec itself.

196 codec round-trip of ground-truth audio scores 0.555, which leaves 0.0747 of headroom above our
197 best system.

198 Training compute is the largest lever: 30k $\rightarrow$ 90k steps gains +0.0827 (CI [0.0733, 0.0908] over 9
199 runs), and a further doubling adds +0.0155. Test-time search is the second lever: best-of- K sampling
200 at matched NFE ($8T$ per candidate), selecting with one speaker encoder and scoring with a second. It
201 gains +0.0444 over refinement at $T=64$ (CI [0.0416, 0.0470], on 72 % of items) and +0.0365 over
202 the deployable $T=16$ default, at a cost of +0.0425 WER. Refinement remains the better spend for
203 intelligibility: WER is 0.1316 under refinement against 0.1955 under search. The gain comes from
204 selection, not from sampling: an arbitrary candidate scores 0.4259 regardless of K . It persists at
205 276 M non-embedding parameters (+0.0422) and, on three runs, at 3 $\times$ training compute (+0.0368).

206 The remaining levers are small or empty. A 14 $\times$ parameter increase gains +0.0531. Training-free
207 speaker-contrastive guidance gains +0.0121 and costs +2.73 WER points, so it trades one capability
208 for the other. Rate-matched length conditioning gains +0.0044, and longer prompts gain nothing:
209 the best prompt is the 3 s default seen in training, and beyond it the output stays fluent and speaker-
210 consistent while decoupling from the text. Identity is bought with training and with selection, not
211 with more refinement.

212 Where the passes are spent matters as much as how many: at matched NFE = 32, moving them to the
213 coarsest codebook level changes WER from 1.1480 to 0.2191, at a cost in identity and naturalness
214 (Appendix D).

Table 6: *Refinement and search buy different capabilities at the same cost.* Matched NFE (8T per candidate); best-of- K candidates are selected by WavLM-SV and scored by a separate ECAPA encoder. Refinement wins intelligibility at every tier, and search wins identity at every tier.

NFE	refinement	search	Δ ECAPA	95% CI	WER (refine / search)
128	$T=16$	$K=2$	+0.0153	[0.0127, 0.0179]	0.1530 / 0.1938
256	$T=32$	$K=4$	+0.0322	[0.0307, 0.0338]	0.1335 / 0.1947
512	$T=64$	$K=8$	+0.0444	[0.0416, 0.0470]	0.1316 / 0.1955
replicated at NFE 512 outside the grid: 276 M budget +0.0422 [0.0393, 0.0448]; $3\times$ training +0.0368 [0.0321, 0.0428]					

Table 7: *The search result is not an artefact of the selector’s lineage.* The same 200 items per run, 15 runs, rescored end-to-end by five speaker encoders spanning four model families. The selector is WavLM-large SV, so its own row is circular and is shown only for reference. A *per-item win rate* — how often an encoder ranks the search output above the refinement output at matched NFE — is invariant to any monotone rescaling of that encoder’s cosine, which a difference of means is not; we therefore compare win rates across encoders and report each raw Δ only beside its own scale. Every family, including the three whose absolute cosines fail our ground-truth gate *purely on scale* while separating speakers at AUC ≥ 0.98 , puts search ahead with an interval clear of one half.

encoder	family / role	win rate	95% CI	raw Δ
ECAPA-TDNN	ECAPA, gated	72.3%	[70.6, 73.8]	+0.0444
x-vector	TDNN	79.0%	[77.5, 80.5]	+0.0075
GE2E d-vector	LSTM d-vector	69.1%	[67.5, 70.8]	+0.0164
WavLM-base+ SV	WavLM, other scale	64.6%	[62.9, 66.3]	+0.0173
WavLM-large SV*	WavLM / <i>selector</i>	84.9%	[83.6, 86.1]	+0.0803

*selects the candidates, so this row is circular and is not counted as agreement.

5 Limitations and outlook

Every model in the grid is under-trained. We cap non-embedding parameters at 133 M and training at 30,000 steps, so T_{WER}^* is a lower bound. Part of the identity gap is also not ours to close: a codec round-trip of the target scores 0.555 against 0.678 for the original waveform, so 62% of the distance to real speech is lost in the codec before the model generates anything.

Our selector and our original scorer share a lineage — both are ECAPA-TDNN-family models, correlating per item at r in the range 0.71–0.73 — so the search result was open to the charge that it measured a shared representational bias rather than identity. Table 7 closes that charge but does not make the measurement lineage-free: the candidates are still *chosen* by one encoder, and what the other four establish is that the chosen candidates are better by their lights too, not that a different selector would choose the same ones. Compute matching also covers the diffusion transformer only, so the selector and scorer forward passes are not charged to the search arm.

The study is one language, one codec and one corpus, with no classifier-free guidance. Target length comes from a corpus median character rate rather than a per-item prediction. Two fitted quantities are not identified: κ sits on its bound in every fit, and the width amplitude saturates its bound, which leaves $\Delta\rho$ undetermined. The $\Delta\tau$ interval reflects seed variation only, and widens to [0.0349, 0.1640] once the fit’s weights are resampled along with the runs.

Table 8 in Appendix C gives every pre-registered verdict. Four hypotheses were refuted outright (H-E1b, H-E2, H-E4b, H-S4), three were not supported (H-D1, H-D3, H-D4) and two were discordant between their pre-specified lenses (H-S1, H-S3); we name them because a paper that quotes only the passing half of a split hypothesis cannot be audited. Two censorings remain, both removable by a larger grid: the depth optimum sits at the deepest shape tested in the two larger budgets, and WER is still falling at the top of the fitted T range.

References

Philip Anastassiou et al. Seed-TTS: A family of high-quality versatile speech generation models. *arXiv preprint arXiv:2406.02430*, 2024.

241 Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured
242 denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing*
243 *Systems (NeurIPS)*, 2021.

244 Zalán Borsos, Matt Sharifi, Damien Vincent, Eugene Kharitonov, Neil Zeghidour, and Marco
245 Tagliasacchi. SoundStorm: Efficient parallel audio generation. *arXiv preprint arXiv:2305.09636*,
246 2023.

247 Huiwen Chang, Han Zhang, Lu Jiang, Ce Liu, and William T. Freeman. MaskGIT: Masked generative
248 image transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*,
249 2022.

250 Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki
251 Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian,
252 Michael Zeng, Xiangzhan Yu, and Furu Wei. WavLM: Large-scale self-supervised pre-training for
253 full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 2022.

254 Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou,
255 Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dialogue.
256 *arXiv preprint arXiv:2410.00037*, 2024.

257 Haorui He, Zengqiang Shang, Chaoren Wang, Xuyuan Li, Yicheng Gu, Hua Hua, Liwei Liu, Chen
258 Yang, Jiaqi Li, Peiyang Shi, Yuancheng Wang, Kai Chen, Pengyuan Zhang, and Zhizheng Wu.
259 Emilia: An extensive, multilingual, and diverse speech dataset for large-scale speech generation.
260 In *IEEE Spoken Language Technology Workshop (SLT)*, 2024.

261 Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza
262 Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al.
263 Training compute-optimal large language models. In *Advances in Neural Information Processing*
264 *Systems (NeurIPS)*, 2022.

265 Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child,
266 Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models.
267 *arXiv preprint arXiv:2001.08361*, 2020.

268 Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-
269 Rong Wen, and Chongxuan Li. Large language diffusion models. *arXiv preprint arXiv:2502.09992*,
270 2025.

271 Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever.
272 Robust speech recognition via large-scale weak supervision. In *International Conference on*
273 *Machine Learning (ICML)*, 2023.

274 Takaaki Saeki, Detai Xin, Wataru Nakata, Tomoki Koriyama, Shinnosuke Takamichi, and Hiroshi
275 Saruwatari. UTMOS: UTokyo-SaruLab system for VoiceMOS challenge 2022. In *Interspeech*,
276 2022.

277 Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T.
278 Chiu, Alexander Rush, and Volodymyr Kuleshov. Simple and effective masked diffusion language
279 models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

280 Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally
281 can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.

282 Chengyi Wang, Sanyuan Chen, Yu Wu, Ziqiang Zhang, Long Zhou, Shujie Liu, Zhuo Chen, Yanqing
283 Liu, Huaming Wang, Jinyu Li, Lei He, Sheng Zhao, and Furu Wei. Neural codec language models
284 are zero-shot text to speech synthesizers. *arXiv preprint arXiv:2301.02111*, 2023.

285 Yuancheng Wang, Haoyue Zhan, Liwei Liu, Ruihong Zeng, Haotian Guo, Jiachen Zheng, Qiang
286 Zhang, Xueyao Zhang, Shunsi Zhang, and Zhizheng Wu. MaskGCT: Zero-shot text-to-speech
287 with masked generative codec transformer. *arXiv preprint arXiv:2409.00750*, 2024.

288 Greg Yang, Edward J. Hu, Igor Babuschkin, Szymon Sidor, Xiaodong Liu, David Farhi, Nick
 289 Ryder, Jakub Pachocki, Weizhu Chen, and Jianfeng Gao. Tensor programs V: Tuning large neural
 290 networks via zero-shot hyperparameter transfer. In *Advances in Neural Information Processing*
 291 *Systems (NeurIPS)*, 2021.

292 Zhen Ye, Xinfu Zhu, Chi-Min Chan, Xinsheng Wang, Xu Tan, Jiahe Lei, Yi Peng, Haohe Liu, Yizhu
 293 Jin, Zheqi Dai, Hongzhan Lin, Jianyi Chen, Xingjian Du, Liumeng Xue, Yunlin Chen, Zhifei Li,
 294 Lei Xie, Qiuqiang Kong, Yike Guo, and Wei Xue. Llasa: Scaling train-time and inference-time
 295 compute for LLaMA-based speech synthesis. *arXiv preprint arXiv:2502.04128*, 2025.

296 A Reproducibility

297 Protocol, hypotheses, thresholds, the frozen sampler and the analysis plan were registered before the
 298 first run, and every gate is logged with its measured values. The released artefact holds the harness,
 299 all 45 run records, the 75-point surface, and the fitting and bootstrap code. Total cost 143 GPU-hours.

300 B The shape grid, as run

301 Every config in the active grid with its measured $T=16$ means (seeds 0, 1, 2, 400 eval_zs items each).
 302 Iso- N within a budget: the three budgets sit at 20, 50 and 125 M non-embedding parameters and the
 303 five shapes per budget span the aspect-ratio range at fixed $N = 12dw^2$.

	config	budget	w	d	heads	N (M)	val loss	WER (%)	SIM-o	Degen (%)
	A1	A	640	4	10	19.7	4.638	53.9	0.327	0.67
	A2	A	448	8	7	19.3	4.582	28.4	0.348	0.08
	A3	A	384	12	6	21.2	4.566	15.9	0.360	0.17
	A4	A	320	18	5	22.1	4.574	13.6	0.359	0.00
	A5	A	256	24	4	18.9	4.611	15.3	0.351	0.17
	B1	B	832	6	13	49.8	4.551	27.8	0.361	0.08
304	B2	B	640	10	10	49.2	4.505	16.0	0.381	0.00
	B3	B	512	16	8	50.3	4.497	11.9	0.388	0.08
	B4	B	448	22	7	53.0	4.511	12.0	0.382	0.08
	B5	B	384	30	6	53.1	4.520	11.5	0.375	0.08
	C1	C	1152	8	18	127.4	4.504	17.1	0.373	0.17
	C2	C	960	12	15	132.7	4.504	18.9	0.373	0.17
	C3	C	768	18	12	127.4	4.488	14.3	0.392	0.00
	C4	C	640	26	10	127.8	4.489	13.4	0.390	0.08
	C5	C	512	36	8	113.2	4.487	9.9	0.388	0.00

305 C Additional figures

306 D Refinement-budget allocation across codebook levels

307 Three frozen per-level step schedules at matched total NFE = 32 on the 125 M budget (15 runs, 400
 308 items). Paired item-level bootstrap, 2000 replicates, pairing on the eval item so item difficulty cancels.
 309 Coarse spends all passes on codebook level 0, fine on level 7, uniform splits evenly.

	schedule	WER	SIM-o	UTMOS	degenerate
310	coarse [25, 1, ..., 1]	0.2191	0.3057	2.17	0.1%
	uniform [4, ..., 4]	0.3144	0.3477	2.51	0.5%
	fine [1, ..., 1, 25]	1.1480	0.2246	1.56	39.1%

311 Allocation is metric-selective in the same way the step *count* is: coarse wins WER but loses identity
 312 and naturalness against uniform, both with intervals excluding zero. It is a trade, not a free win.

Table 8: *The complete pre-registration ledger*. Every hypothesis declared before the first run, with its verdict — including the four that failed and the two that split. We report the failures beside the successes because a ledger that shows only its winners cannot be used to judge the winners. DISCORDANT is a declared outcome class, used where the two pre-specified lenses disagree, and is not a synonym for refuted.

id	claim	verdict
H-D1	fixed- T anisotropy $\Delta\rho > 0$	not supported
H-D2	$\Delta\tau > 0$, metric-selective steps	supported [§]
H-D3	depth-step exchange rate κ	not supported
H-D4	small budgets predict the largest	not supported
H-E1a	$T_{\text{WER}}^* > 16$	supported
H-E1b	extended τ inside the v1.0 CI	refuted
H-E2	iso-latency allocation is depth-first	refuted
H-E3	coarse-level NFE beats fine	supported
H-E4a	scale persistence, four budgets	supported
H-E4b	predicted vs observed within 15 %	refuted
H-E5	survives $3\times$ training compute	supported
H-S1	longer prompt context buys identity	discordant
H-S2	search buys identity	supported
H-S3	rate-matched length buys identity	discordant
H-S4	contrastive guidance buys identity	refuted

[§]supported as declared; see §4.2 for the coordinate, range and fit conditions that bound it.

Table 9: *Inference-time interventions that do not buy identity*. Speaker-contrastive guidance trades intelligibility for similarity and degrades monotonically in γ ; per-item rate matching gives a small mean gain that is not reliable item-by-item.

intervention	setting	Δ SIM-o	95% CI	Δ WER (pts)
contrastive guidance	$\gamma = 0.5$	+0.0121	[0.0094, 0.0150]	+2.73
	$\gamma = 1.0$	+0.0025	[-0.0015, 0.0063]	+7.22
	$\gamma = 2.0$	-0.0278	[-0.0327, -0.0231]	+17.26
rate matching	per-item duration	+0.0044	[0.0017, 0.0067]	+0.86
13/15 runs positive, but a per-item sign test gives $p = 0.051$				

Table 10: *Longer prompts do not buy identity, and destroy intelligibility*. Prompt length swept at inference on 15 125 M-budget runs, 397 items, every arm scored against the same fixed 3 s reference. Training used 3 s prompts only, so the 6 and 9 s arms test extrapolation. There the output stays fluent and speaker-consistent, but it decouples from the conditioning text.

prompt	SIM-o	WER
1.5 s	0.3213	0.1578
3 s (<i>training length</i>)	0.3839	0.1474
6 s	0.3724	0.7692
9 s	0.3759	0.8642

A significance-based saturation contrast. Defining saturation as the smallest T whose paired-bootstrap interval for $\text{err}(T) - \text{err}(64)$ covers zero gives $T_{\text{WER}}^* = 32$ against $T_{\text{SIM}}^* = 16$ on the extended grid. We report this only here and do not build on it: roughly half the contrast is a precision artefact, since identity’s detection floor is $\approx 1.8\times$ looser relative to its own dynamic range than intelligibility’s. The raw-scale T^* — 95% of the $T=16$ value — is not affine-invariant and must never be compared across metrics: the same 5% band is 0.67% of WER’s range but 11.96% of SIM-o’s. On the invariant fraction-of-total-gain definition, $T^* = 16$ for both metrics.

E Fit diagnostics and exploratory analyses

Parameters at bounds. The width amplitude A saturates the pre-registered *upper* bound $A \leq 10$ in every fit (Part A: 10.000 for both metrics; Part B: 9.999 for WER, 10.000 for SIM-o), so only the product $A w^{-\alpha}$ is identified and α is not; the bound, not the data, is binding, which is a limitation

Table 11: *Where the refinement passes are spent matters as much as how many.* Three frozen per-level schedules at matched total NFE = 32 on the 125 M budget, 15 runs, 400 items. Spending every pass on the coarsest codebook level rather than the finest cuts WER by $5.2\times$, and costs identity and naturalness against a uniform split.

schedule	WER	SIM-o	UTMOS	degenerate
coarse [25, 1, . . . , 1]	0.2191	0.3057	2.17	0.1%
uniform [4, . . . , 4]	0.3144	0.3477	2.51	0.5%
fine [1, . . . , 1, 25]	1.1480	0.2246	1.56	39.1%

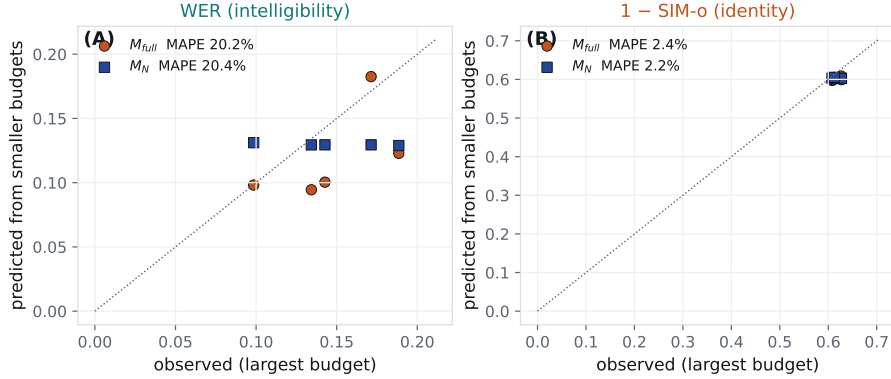


Figure 5: *H-D4 extrapolation.* The two smaller budgets are fitted and the largest budget predicted, under M_{full} and the N -only model. SIM-o extrapolates (MAPE 2.4 % vs 2.2 %); WER does not (20.2 % vs 20.4 %).

of the pre-registered parameterisation worth fixing in any follow-up. κ reaches its upper bound in every one of the 2000 bootstrap replicates of M_{sub} . The irreducible term E reaches 0 for WER under M_{sep} . The depth amplitude B is interior throughout, so β is identified.

Exploratory: the depth exponent separates the capabilities. $\beta_{WER} = 2.040$ against $\beta_{SIM} = 0.553$, a factor of 3.7 — the asymmetry H-D1 sought, in the parameter the data pin down. Part-B ρ values point the pre-registered way ($\Delta\rho = +0.890$) but inherit the A -cap problem; the pre-registered test is Part A, so this is exploratory only.

Weighting. Points are weighted $1/SE^2$ with SE taken over seeds, floored at a quarter of the pooled seed SD so that a chance agreement between two seeds cannot dominate the fit; the number of floored points is recorded in the released `fits.json`.

Exploratory (not part of any declared claim). An unweighted refit of both parts is released alongside the weighted fit, as is a fit that additionally includes the finished runs of a stretch third seed. Both reproduce the sign and significance of $\Delta\tau$.

Use of Generative AI Disclosure

Generative AI was used as a coding assistant for the experimental harness and for language refinement of author-written text. The research design, the pre-registered hypotheses and thresholds, the analysis plan, and all scientific claims are the authors' own; the authors verified the code and the numbers and take full responsibility for this paper.